

Finding missing values using Random Forest

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	??	??	No

Initial guess for Block Arteries = No (most frequent)

Initial guess for Weight = $\text{Avg}(125, 210) = 167.5$

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
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Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	No	167.5	No

Build a Random Forest using the data and run every example through each of the decision tree

Suppose in 1st tree example 3 and 4 end up together in a leaf

Suppose in 2nd tree example 2, 3 and 4 end up together in a leaf

Suppose in 3rd tree example 3 and 4 end up together in a leaf

and so on

	Ex1	Ex2	Ex3	Ex4
Ex1				
Ex2			1	1
Ex3		1		1+1
Ex4		1	1+1	

After running through 10 trees the proximity matrix looks like

	Ex1	Ex2	Ex3	Ex4
Ex1		2	1	1
Ex2	2		1	1
Ex3	1	1		8
Ex4	1	1	8	

Dividing by 10

	Ex1	Ex2	Ex3	Ex4
Ex1		0.2	0.1	0.1
Ex2	0.2		0.1	0.1
Ex3	0.1	0.1		0.8
Ex4	0.1	0.1	0.8	

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Yes	No	??	??	No

Weighted frequency of Yes = $\frac{1}{3} \times \{0.1/(0.1+0.1+0.8)\} = 0.03$

Weighted frequency of No = $\frac{2}{3} \times \{(0.1+0.8)/(0.1+0.1+0.8)\} = 0.6$

The missing value is “No”

Weighted avg weight = $125 \times \{0.1/(0.1+0.1+0.8)\} + 180 \times \{0.1/(0.1+0.1+0.8)\} + 210 \times \{0.8/(0.1+0.1+0.8)\} = 198.5$

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Yes	Yes	No	210	No
Yes	No	No	198.5	No

Build a random forest, run all the examples through the all the trees, build the proximity table and update the missing values

This procedure will continue until the values converge

Classify No, No, No, ??,

Assume No, No, No, ??, No and No, No, No, ??, yes

Find missing values separately and classify both examples using the random forest.

If more trees predict yes then class is yes

If more trees predict no then class is no